INCH-POUND
MIL-S-24187/8A(SH)
22 March 1991
SUPERSEDING
MIL-S-24187/8(SH)
9 September 1983

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, REMOTE OPERATION, TYPE R3DLS0-S3A3 MANUAL AND REMOTE OPERATION, STYLE DLS, TYPE R3DLS0-S3A4

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

INCHES	MM
.001	0.03
.005	0.13
.03	0.76
.12	3.05
.28	7.11
.312	7.93
.45	11.43
3.060	77.72
4.00	101.63
7.060	179.32
7.96	202.18
9.40	238.76
9.96	252.98
18.00	457.20
20.67	525.02
23.20	589.28

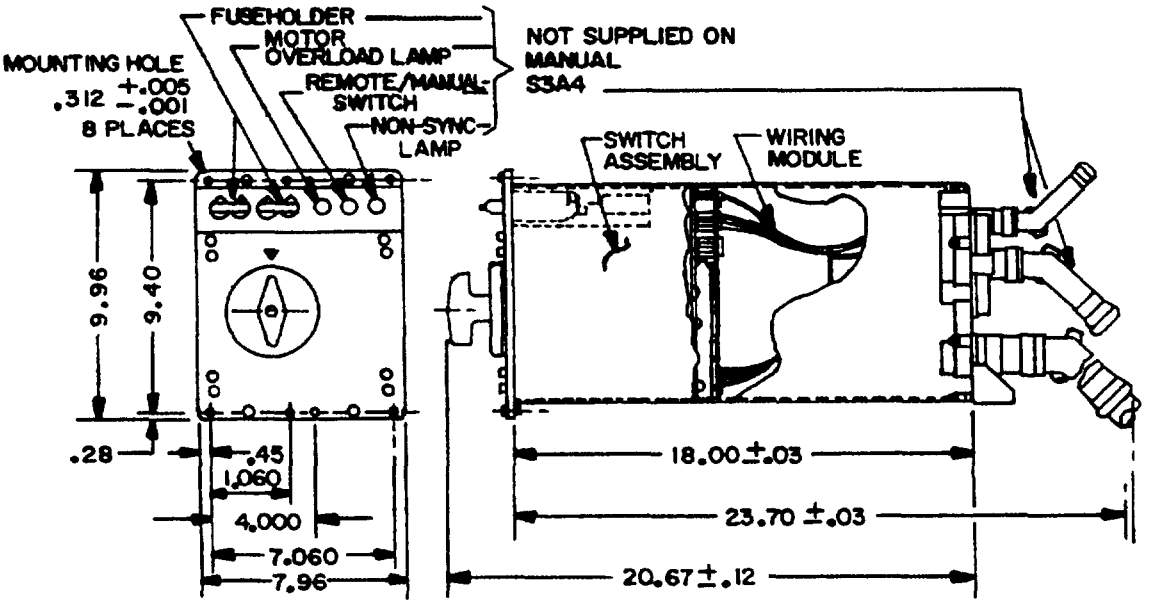


FIGURE 1. *Example Type R3DLS0-S3A3 panel mount switch assembly.*

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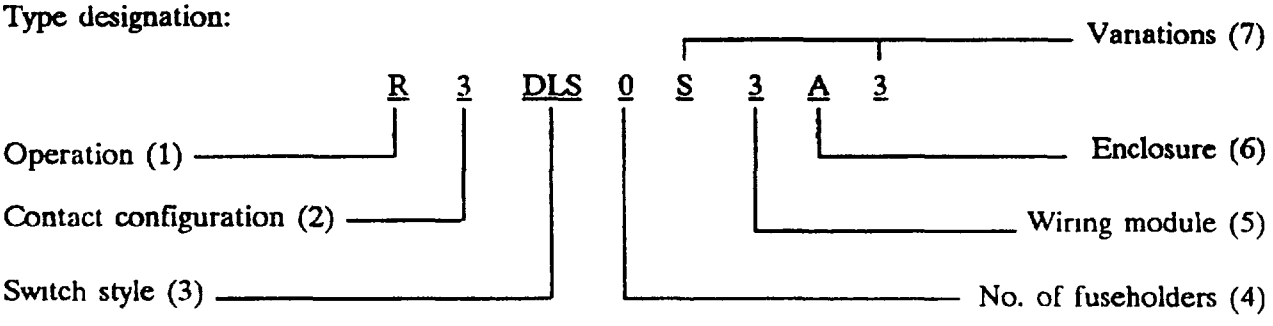
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NOTES:

1. Unless otherwise specified dimensions are in inches, tolerances plus or minus 0.02 (0.51 mm) for two-place decimals and plus or minus 0.005 (0.13 mm) for three-place decimals.
2. Connectors plugs with 45 degree adapters shall be supplied unless otherwise specified (See figures 6A and 6B.)
   6 pcs 92 pin plug, MIL-C-28840 (M28840/16AG1S1) with M28840/916WA Back shell -S3A3 and S3A4
   1 pc 14 pin plug, MIL-C-5015 (MS3406D20-27S) with M85049/6-16W Back shell -S3A3 only.
   1 pc 2 pin plug, MIL-C-5015 (MS3406D12S-3S) with M85049/6-6W Back shell -S3A3 only.
   1 pc 12 pin plug, MIL-C-28840 (M28840/16AB1S1) with M28840/903WA - Remote S3A4 only.
3. For suggested mounting see Drawing 815-5002992.
4. Metric equivalent (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. *Example: Type R3DLSO-S3A3 panel mount switch assembly – Continued.*

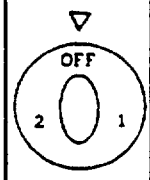
Type designation:



(1) Switch assemblies shall be remote operated with manual override, designated R, or manually operated, designated M (S3A4 only).
(2) Contact configuration shall be type 3 (see figure 2).
(3) Switch type shall be designated by the letters 3DLS0-S3A3, or 3DLS0-S3A4
(4) There shall be no provisions for additional fuseholders.
(5) Wiring module circuitry shall be indicated by the designator 3 as specified to accomplish the switching action specified in table I.
(6) Enclosed switch assemblies shall be indicated by a single letter (A)
(7) Shipboard connectors, latching relay circuit (remote only) and wiring configuration are indicated by designations S & 3, or S & 4

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TABLE I. *Switch selection.*

Type designation						Application data					
Switch operation	Contact configuration	Switch style	Additional fuseholders	Wiring module with enclosure	Switching action[1]	Stop position	Dial position[2]	Handle position[3]	Wired contacts[4]	To ship wiring connection[5]	Unwired contact[4]
R	3	DLS	0	-S3A3	1-OFF-2	2&7			( )C LOAD 2 ( )A LOAD 1 ( )X INPUT	JE, JF JF, JE JC, JD, JG, JH	( )D ( )B
M or R	3	DLS	0	-S3A4					( )C LOAD 2 ( )A LOAD 1 ( )X INPUT	JG, JH JE, JF JC, JD	

Example. Type designation for a 1-OFF-2 switch assembly with enclosure, R3DLS0-S3A3

[1]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated.

[2]Reference position of the switch is considered to be that in which the flat on the handle shaft is up. The mark ∇ and the dial positions shown are used to designate the reference position.

[3]Normal position of the switch is the position which coincides with normal circuit conditions in any application. Handle shall be adjustable to point upward. The handle and rotary description plate rotate.

[4]See figures 2 and 3 for complete description of contact identification and table IV for load switching.

[5]See figures 4 through 7 and tables IV and V for connector identifications and arrangements.

REQUIREMENTS:

Construction/wiring: Switch parts, modules and assemblies shall be constructed and wired in accordance with the applicable drawing:

815-5002992 for type R3DLS0-S3A3
803-5727191 for type R/M 3DLS0-S3A4

Characteristics: Characteristics shown in table II shall apply to the type R3DLS0-S3A3 and S3A4, switch assemblies.

Drip proof integrity of the enclosure shall be maintained in accordance with Drawings 815-5002992 for -S3A3 and 803-5727191 for -S3A4.

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TABLE II. *Characteristics.*

Characteristics	Value
Total number of poles available	210
Wired load switching poles	160
Number of positions	3
Break-before-make	Yes
Make-before-break	No
Electrical rating and endurance loading	
2 Amperes, 15 Vdc	20,000 Operations
25 Microamperes, 1.5 Vdc	20,000 Operations
Remote operation.	
Transfer time:	
Minimum	0.010 seconds
Maximum	0.035 seconds
Actuating time.	
1 Position change	0.25 seconds
2 Position change (one direction)	0.60 seconds
Current drain:	
At 115 V, 3 ∅, 60 Hz	1 Ampere rms per phase
Contact resistance	
At 2 amperes	0.025 ohm
At 2.5 milliamperes	0.05 ohm
Total circuit resistance:	
At 2 amperes	0.065 ohm
At 25 microamperes	0.100 ohm
Torque – shaft rotation maximum – Initial	90 pound-inches
– After shock	100 pound-inches
– stop screw (strength)	250 pound-inches
Maximum weight	
Enclosed with wiring module	40 pounds
Without enclosure and wiring module	30 pounds
Temperature and humidity (for rotational torque and endurance tests)	
Minimum	Plus 40 °F (plus 4 °C)
Maximum	Plus 149 °F (plus 65 °C) and 40 to 50 percent rh
Shock	MIL-S-901, grade A, class I, type C
Vibration	MIL-STD-167-1, type I, except omit variable frequency
Temperature rise	30 °C

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Switch assembly modules: Switch assemblies shall consist of the following modules as detailed on Drawings 815-5002992 for -S3A3 or 803-5727191 for -S3A4.

Switch assembly
- Front plate module
- Mechanism module
- Contact module

Wiring module
Enclosure

Remote operated switch assemblies shall have in addition to the above, the Remote Control Module (associated circuitry) Control Circuit Module.

Switch function: Switch assemblies shall provide the number of poles indicated - each consisting of five stationary contacts X, A, B, C and D and a movable contact arranged as shown on figure 2. A stop (screw) shall prevent transfer of the movable contact into position X. Stops are referenced on Drawings 815-5002992 for -S3A3 or 803-5727191 for -S3A4.

For a 1-OFF-2 operation, the stops shall be located to prevent the switch from going to the X and D positions. Load 1 shall be connected to contact A, contact B is not connected and load 2 shall be connected to contact C.

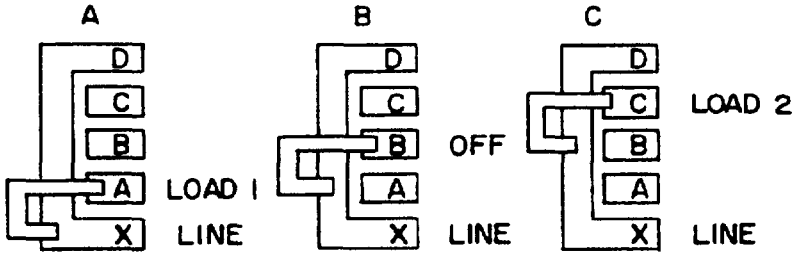


FIGURE 2. *Schematic typical operation of 1 pole.*

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TABLE III. *Stop positions versus switching action.*

Stop position	Switch positions accessible	Switching action
2 and 7	A-B-C	1-OFF-2

Contact designation marking: Each stationary switching contact shall be identified by a number, two and letters and a second number. The first number shall identify the switch section in which the contact is located. The first letter, A, B, C, D, E, F, or G shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figures 2 and 3. The second number shall be that of the column in which a contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects. Contact D is common with contact X in all poles. Contacts shall be installed as shown on figure 3. Circles signify omitted contacts, phantom lines indicate mating header plate.

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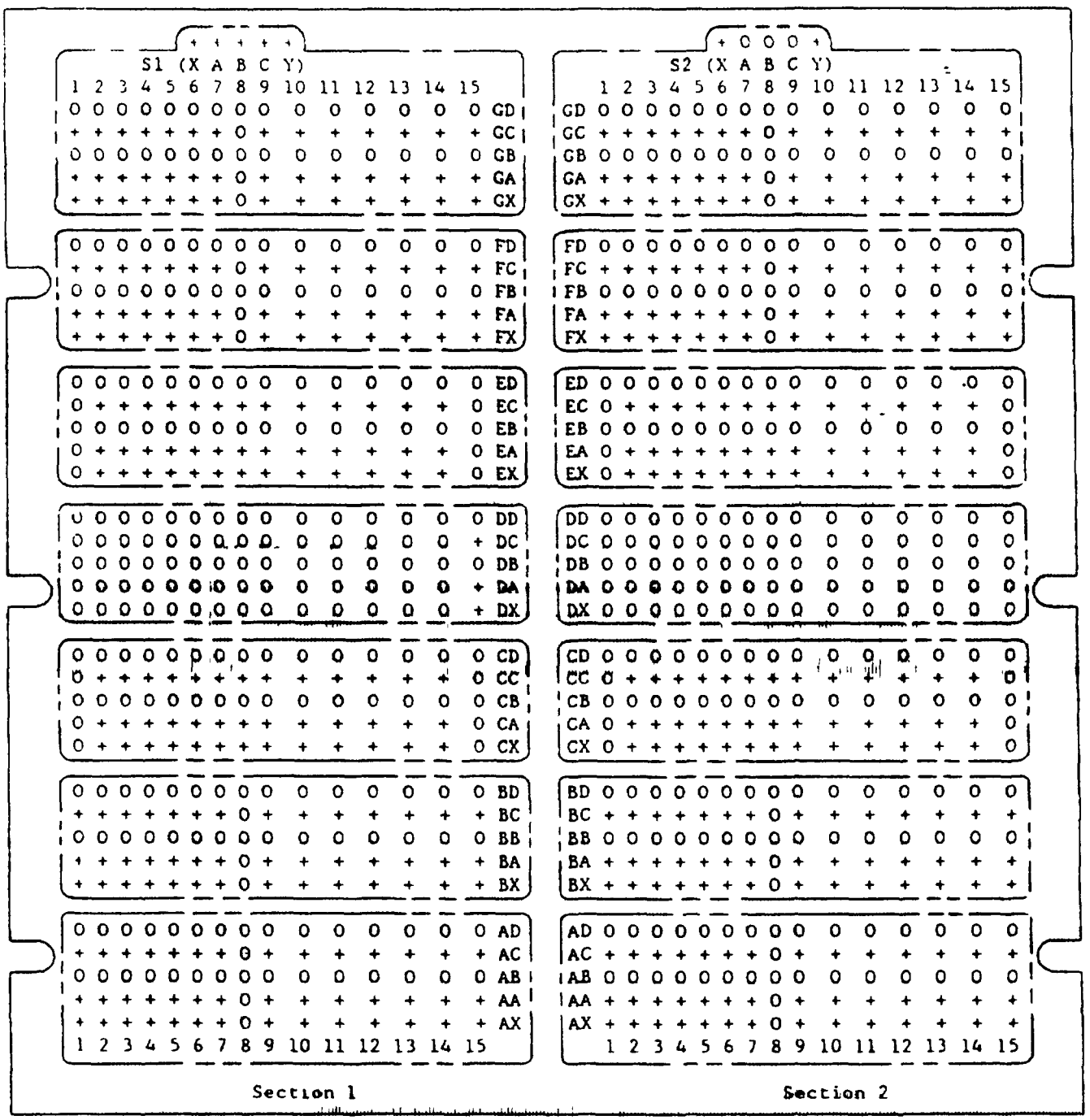


FIGURE 3. *Contact designation and marking, rear view.*

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Enclosure and wiring switch module: Enclosure shall be furnished with the wiring module to connect the contact module to the external mating connectors at the rear of the enclosure (see figures 6A and 6B).

The general arrangement of the wiring shall be as shown on figures 4A and 5 for type -S3A3 and figures 4C and 5 for type -S3A4.

The interconnecting wiring shall be as shown in tables IV and V. The connector contact locations are shown on figure 7A for type -S3A3 and figure 7B for type -S3A4.

If shield switching is required and specified, the shield circuit wiring shall be as shown on figure 4B for type -S3A3 and figure 4D for type -S3A4.

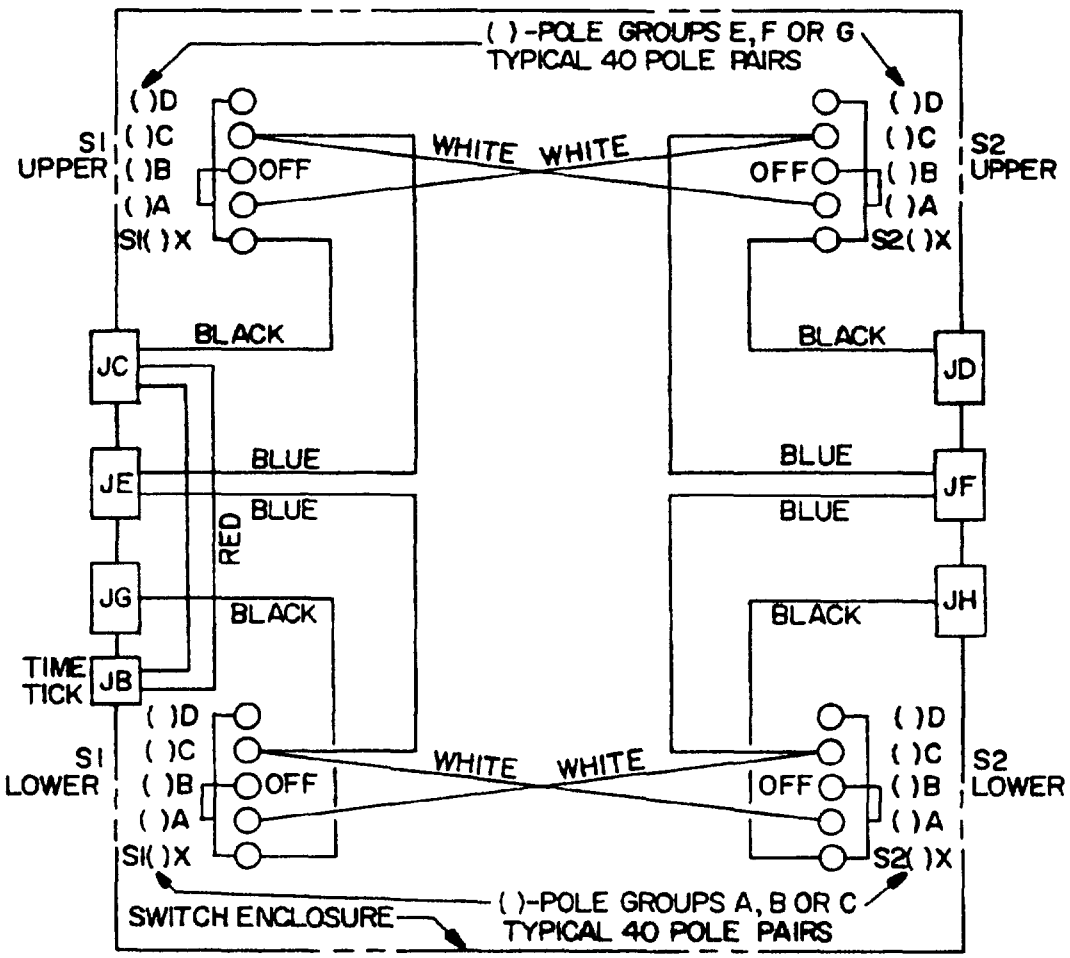


FIGURE 4A. *Interconnecting wiring arrangement. Type -S3A3*

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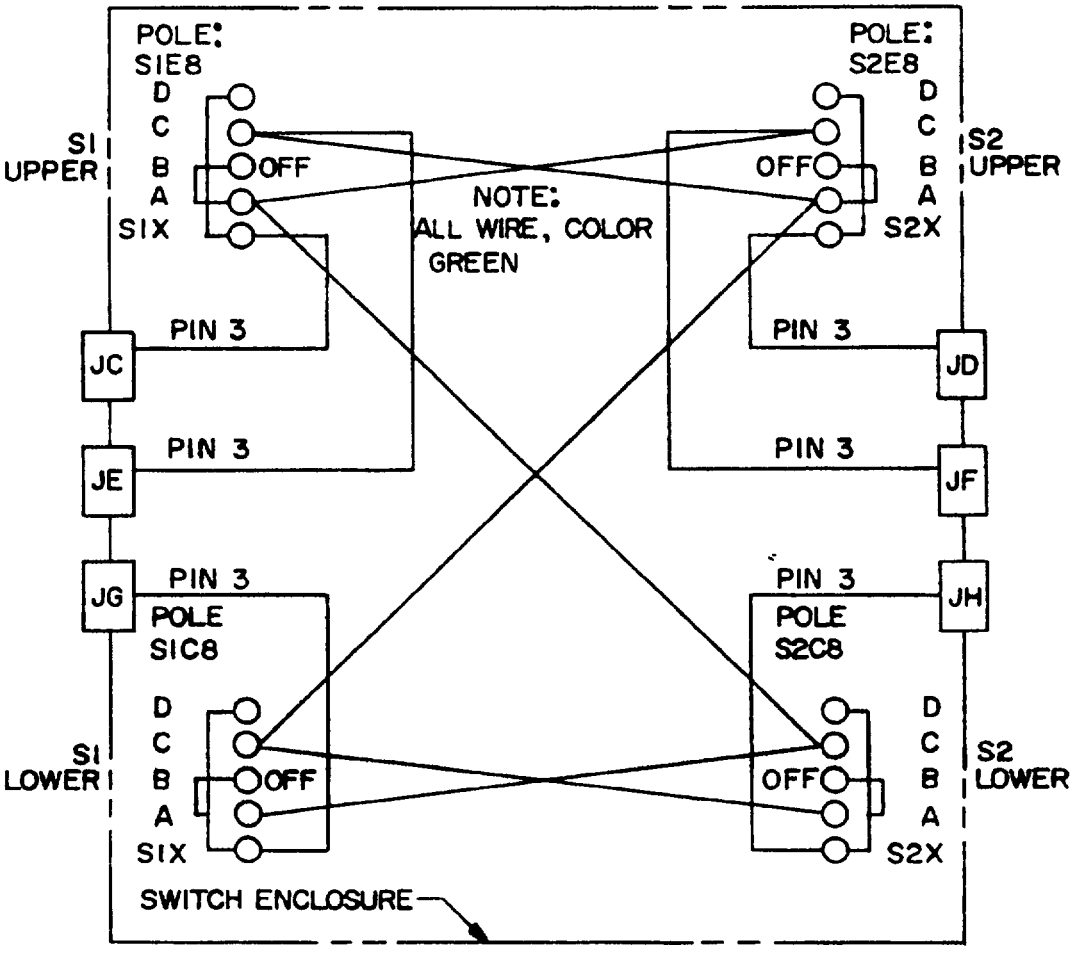


FIGURE 4B. *Shield circuit wiring.*
Type -S3A3

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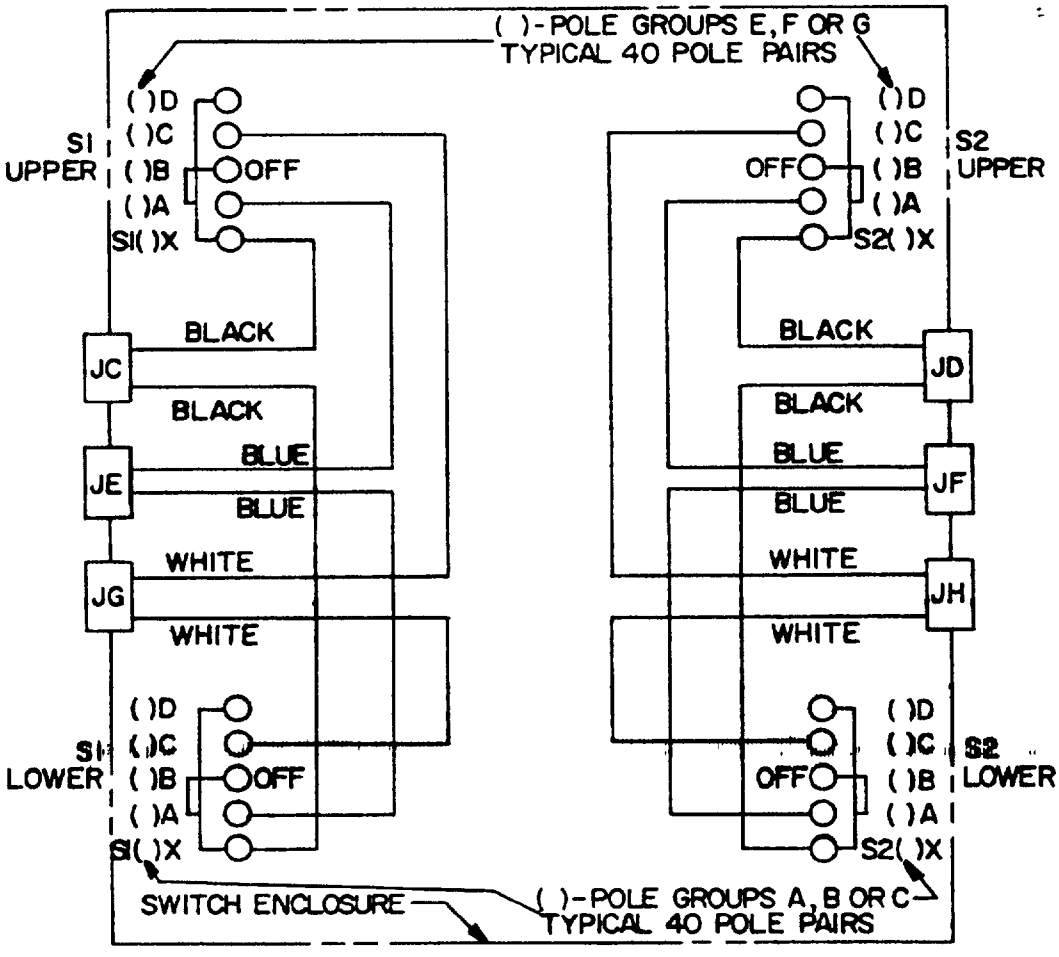


FIGURE 4C. *Interconnecting wiring arrangement. Type -S3A4*

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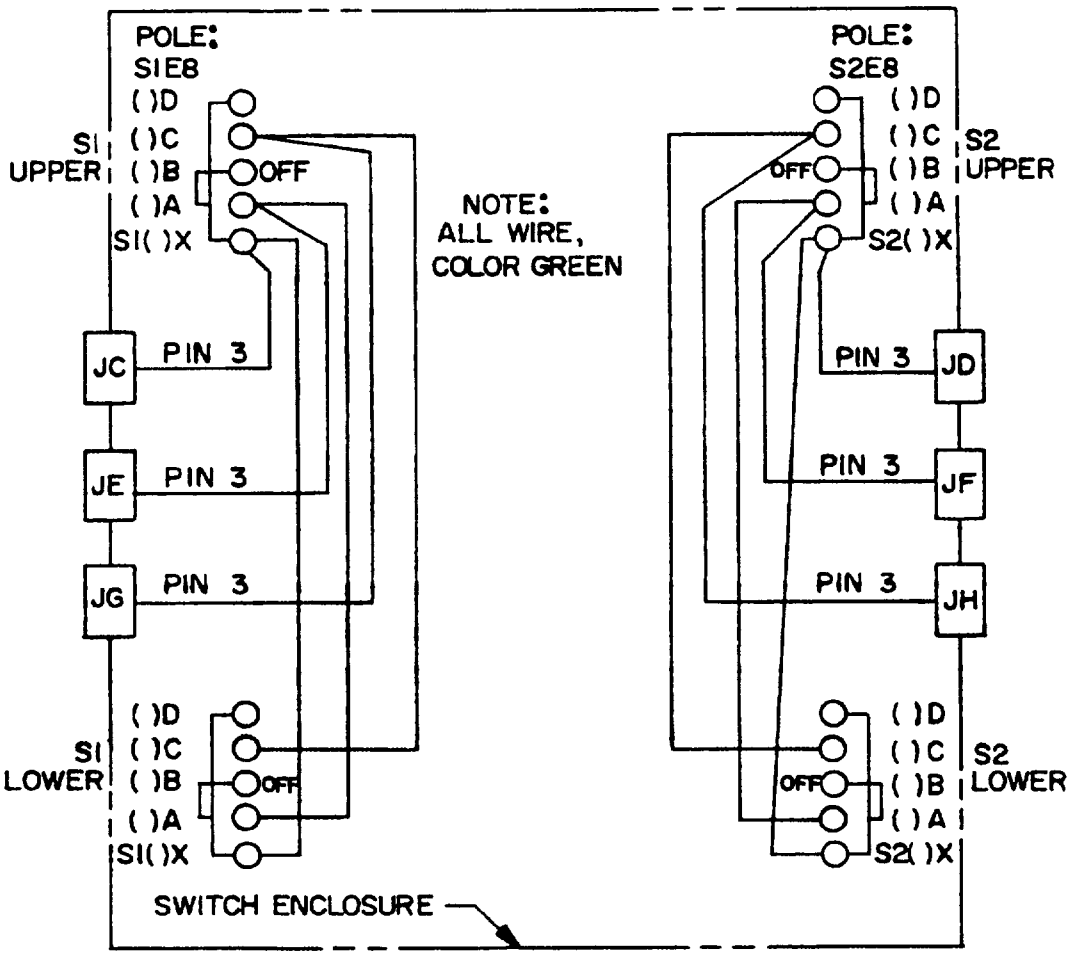


FIGURE 4D. *Shield circuit wiring.*
Type -S3A4

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TABLE IV. *Wiring module connections (rear view).*

S3A4		S3A3		Section 1 Switch module contacts														Section 2 Switch module contacts															S3A3		S3A4
92 Pin Conn	Pole Pairs	92 Pin Conn	Column Pole	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Column Pole	92 Pin Conn	Pole Pairs	92 Pin Conn
				13	21	12	5	11	4	10		9	2	8	1	7	15	6	13	21	12	5	11	4	10		9	2	8	1	7	15	6				
JC		JC	G	22	31	30	20	29	19	28		18	27	17	26	16	25	24	22	31	30	20	29	19	28		18	27	17	26	16	25	24	G	JD		JD
JE	Upper	and	F		32	41	40	39	38	37	3	36	35	34	33	23	14			32	41	40	39	38	37	3	36	35	34	33	23	14		F	and	Upper	JF
JG	40	JE	E																														E	JF	40	JH
			D		70	60	59	58	57	56	3	55	54	53	52	61	71			70	60	59	58	57	56	3	55	54	53	52	61	71		D			
JC		JG	C	79	69	68	77	67	76	66		75	65	74	64	73	63	62	79	69	68	77	67	76	66		75	65	74	64	73	63	62	C	JH		JD
JE	Lower	and	B	87	78	86	92	85	91	84		83	89	82	88	81	72	80	87	78	86	92	85	91	84		83	89	82	88	81	72	80	B	and	Lower	JF
JG	40	JE	A	92 Pin connector contacts																													A	JF	40	JH

EACH 92 PIN CONNECTOR FOR THE LOAD SWITCHING CIRCUITS SHALL BE WIRED TO A SPECIFIC GROUP OF CONTACTS OF A SWITCH CONTACT MODULE USING THE ABOVE TABLE IV

FOR S3A3

EXAMPLE. UPPER 40 POLE PAIRS (GROUP E, F AND G)

POLE E5 CONTACT X SECTION 1 IS WIRED TO PIN 39 OF CONNECTOR JC USING BLACK WIRE
POLE E5 CONTACT C SECTION 1 IS WIRED TO PIN 39 OF CONNECTOR JE USING BLUE WIRE AND TO POLE E5 CONTACT A SECTION 2 USING A WHITE JUMPER WIRE
POLE E5 CONTACT X SECTION 2 IS WIRED TO PIN 39 OF CONNECTOR JD USING BLACK WIRE
POLE E5 CONTACT C SECTION 2 IS WIRED TO PIN 39 OF CONNECTOR JF USING BLUE WIRE AND TO POLE E5 CONTACT A SECTION 1 USING A WHITE JUMPER WIRE
POLE E5 CONTACT D IS UNWIRED AND COMMON TO CONTACT X

EXAMPLE: LOWER 40 POLE PAIRS (GROUP A, B AND C)

POLE C5 CONTACT X SECTION 1 IS WIRED TO PIN 58 OF CONNECTOR JG USING BLACK WIRE
POLE C5 CONTACT C SECTION 1 IS WIRED TO PIN 58 OF CONNECTOR JE USING BLUE WIRE AND TO POLE C5 CONTACT A SECTION 2 USING A WHITE JUMPER WIRE
POLE C5 CONTACT X SECTION 2 IS WIRED TO PIN 58 OF CONNECTOR JH USING BLACK WIRE
POLE C5 CONTACT C SECTION 2 IS WIRED TO PIN 58 OF CONNECTOR JF USING BLUE WIRE AND TO POLE C5 CONTACT A SECTION 1 USING A WHITE JUMPER WIRE
POLE C5 CONTACT D IS UNWIRED AND COMMON TO CONTACT X

PIN 3 OF EACH 92 PIN CONNECTOR, SHALL BE USED FOR THE SHIELDED WIRE CIRCUIT USING GREEN WIRE AND JUMPERS BETWEEN POLES C8 AND E8 FOR BOTH SECTIONS 1 AND 2, THIS APPLIES FOR BOTH S3A3 AND S3A4

FOR S3A4

EXAMPLE. UPPER 40 POLE PAIRS (GROUP E, F AND G)

POLE E5 CONTACT X SECTION 1 IS WIRED TO PIN 39 OF CONNECTOR JC USING BLACK WIRE
POLE E5 CONTACT A SECTION 1 IS WIRED TO PIN 39 OF CONNECTOR JE USING BLUE WIRE
POLE E5 CONTACT C SECTION 2 IS WIRED TO PIN 39 OF CONNECTOR JG USING WHITE WIRE
POLE E5 CONTACT D IS UNWIRED AND COMMON TO CONTACT X

EXAMPLE: LOWER 40 POLE PAIRS (GROUP A, B AND C)

POLE C5 CONTACT X SECTION 2 IS WIRED TO PIN 58 OF CONNECTOR JD USING BLACK WIRE
POLE C5 CONTACT A SECTION 2 IS WIRED TO PIN 58 OF CONNECTOR JF USING BLUE WIRE
POLE C5 CONTACT C SECTION 2 IS WIRED TO PIN 58 OF CONNECTOR JH USING WHITE WIRE
POLE C5 CONTACT D IS UNWIRED AND COMMON TO CONTACT X

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TABLE V. *Wiring module connections.*

	Connector	S3A3 Contact	S3A4 Contact	Terminal board contact	Switch control module contact
				TB	
	JA	A	1	TB1-1	S1A to S2X
		B	4	TB1-2	S1B to S2Y
		C	10	TB1-3	S1C to S1DX15
		Not used	11	TB1-5	S1X
		D,E,M,N	9	- NOT USED -	
		F	12		S1Y
		G	8	TB2-1	–
		H	7	TB2-2	–
		I	3	TB2-3	–
		J	2	TB2-4	–
		K	6		S1DA15
		L	5		S1DC15
Type R3DLSO-S3A3 only	JB	A to		–	–
	JC	50		–	–
	–	–	–	TB1-5	S1X
	JB	B to		–	–
	JC	51		–	–

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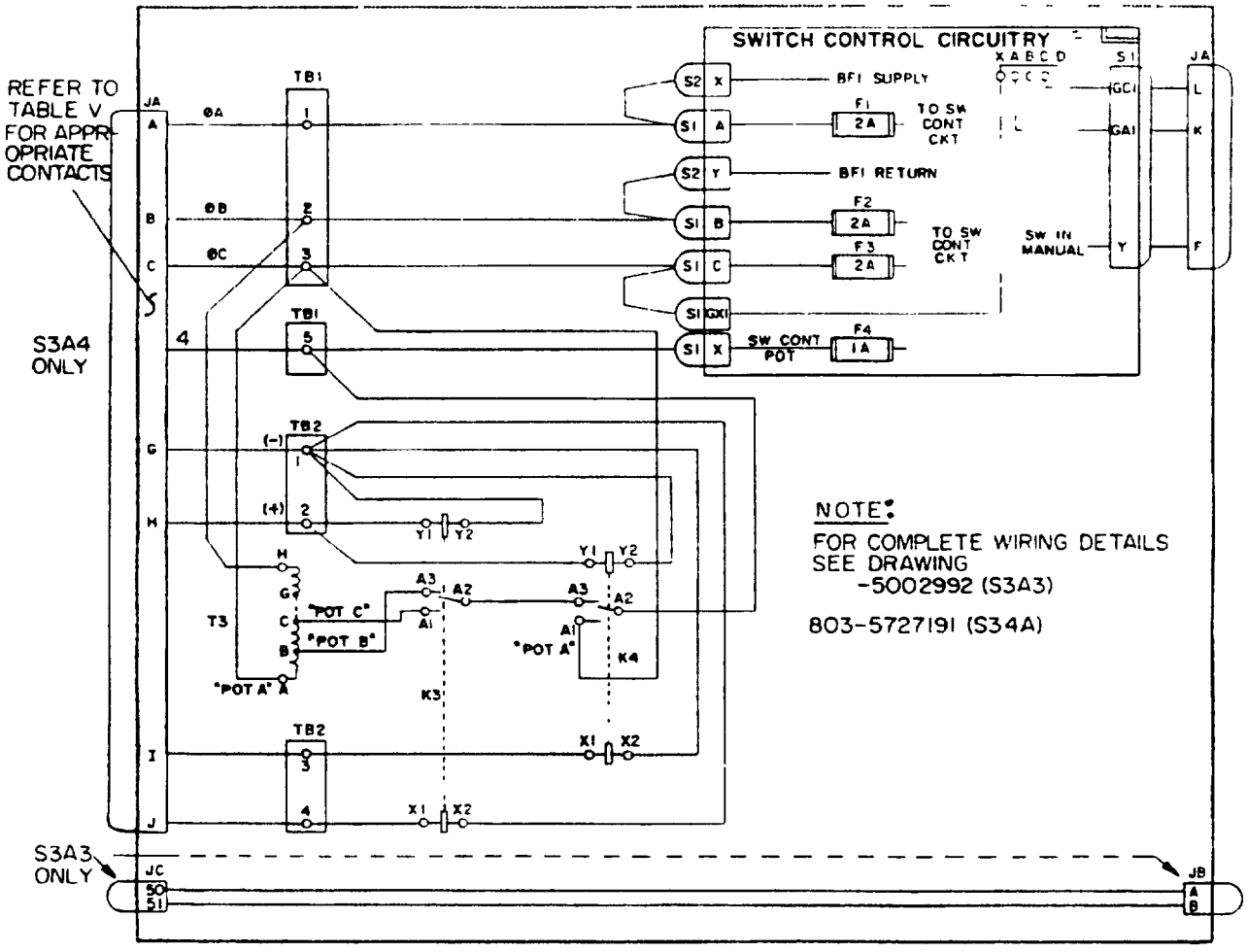


FIGURE 5. *Wiring schematic; switch control circuitry and latching relay circuit portion of wiring module*

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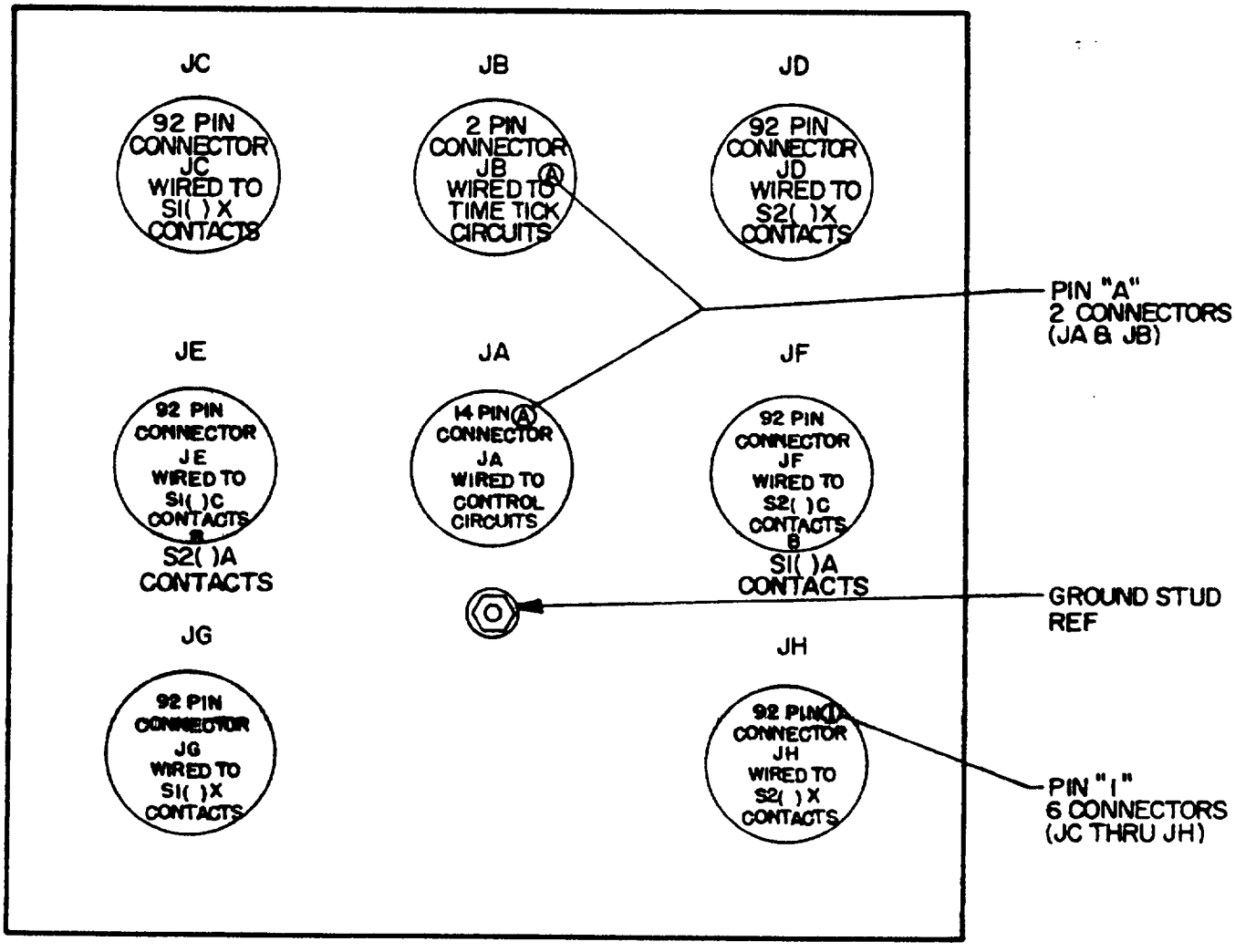


FIGURE 6A. *Enclosed R3DLS0-S3AS switch assembly connector arrangement, rear view.*

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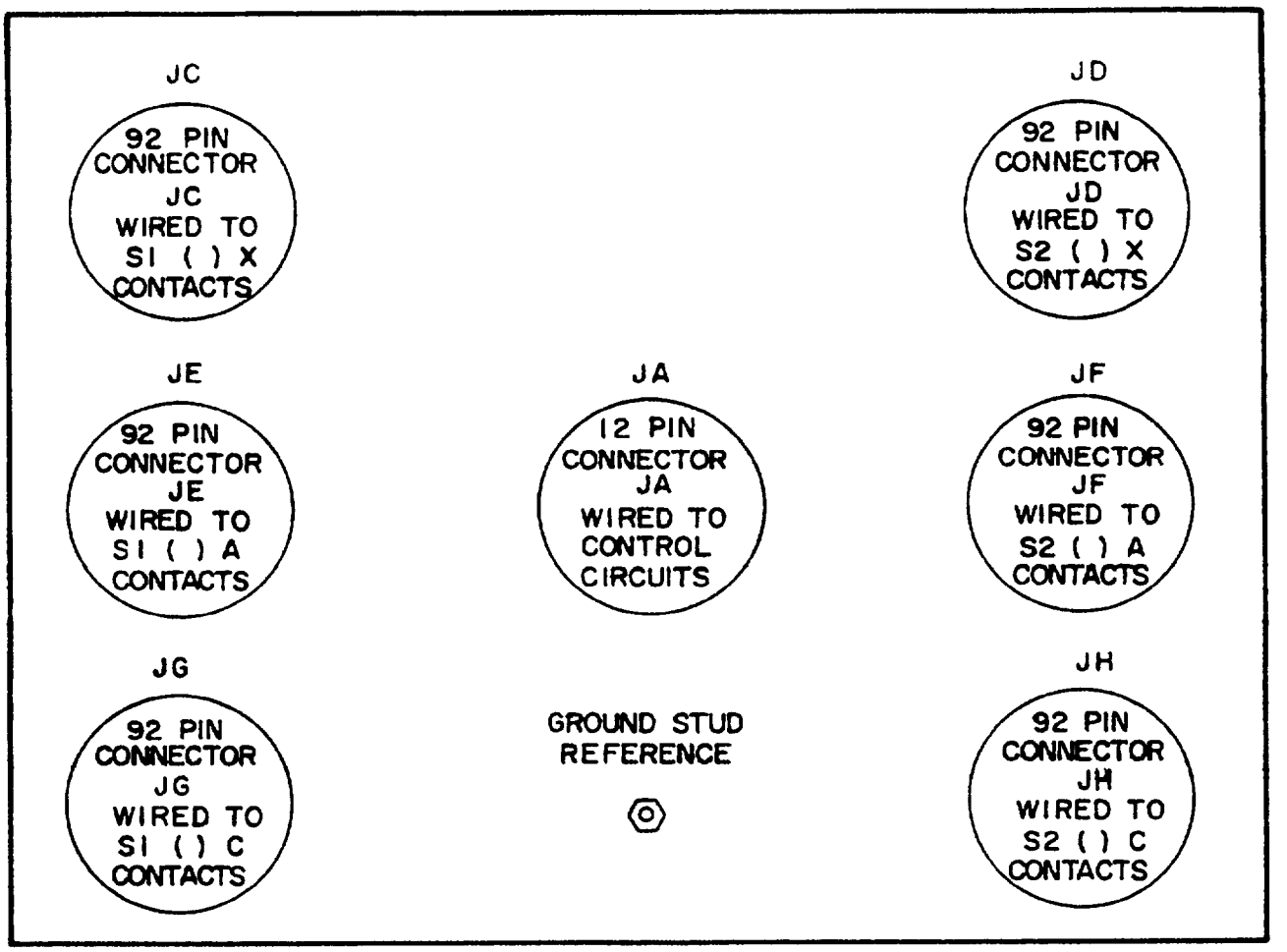


FIGURE 6B. *Enclosed R3DSL0-S3A4 switch assembly connector arrangement, rear view.*

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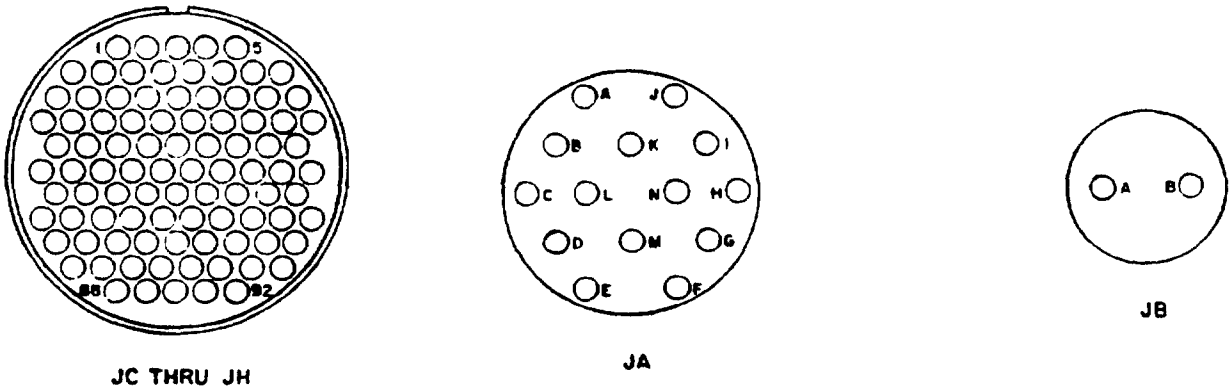


FIGURE 7A. *R3DLS0-S3A3 connector contacts location as viewed from wire insertion face.*

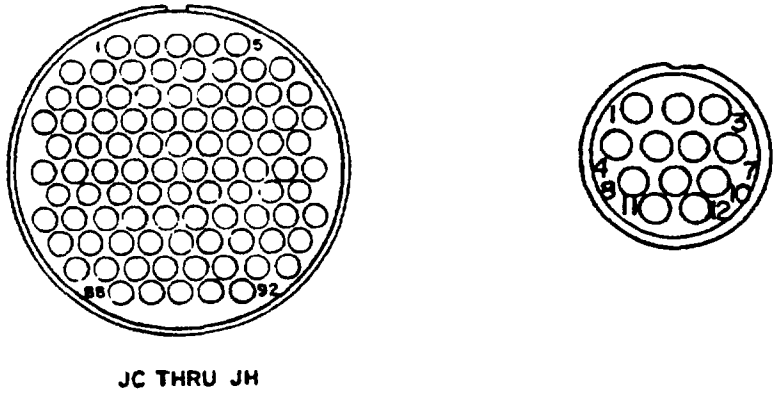
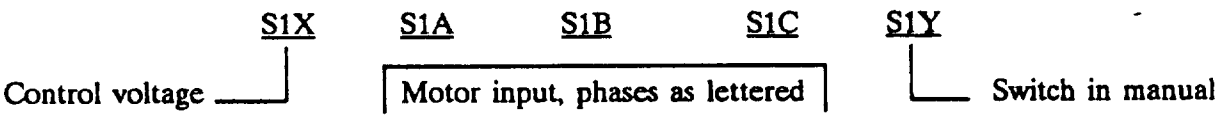


FIGURE 7B *R3DLSL0-S3A4 connector contacts location as viewed from wire insertion face.*

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Remote operation: Connections to the switch module for remote operation shall be made to section 1 contacts as follows:



Remotely operated switch assembly shall move to the positions shown in table VI when the specified control voltage is applied.

TABLE VI. *Control voltage for remote operation.*

Control voltage	Closed switch contacts	XFMR tap[1]
	1-OFF-2	
0 16.4 32.8 49.2	X-A X-B (off) X-C Not used	A B C D

[1]See figure 3 of MIL-S-24187 for typical test circuits for remote operated switch assemblies. Test loading is shown on figure 9 of this document.

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Remote control circuit: Control circuit for remote operation shall be in accordance with figure 8.

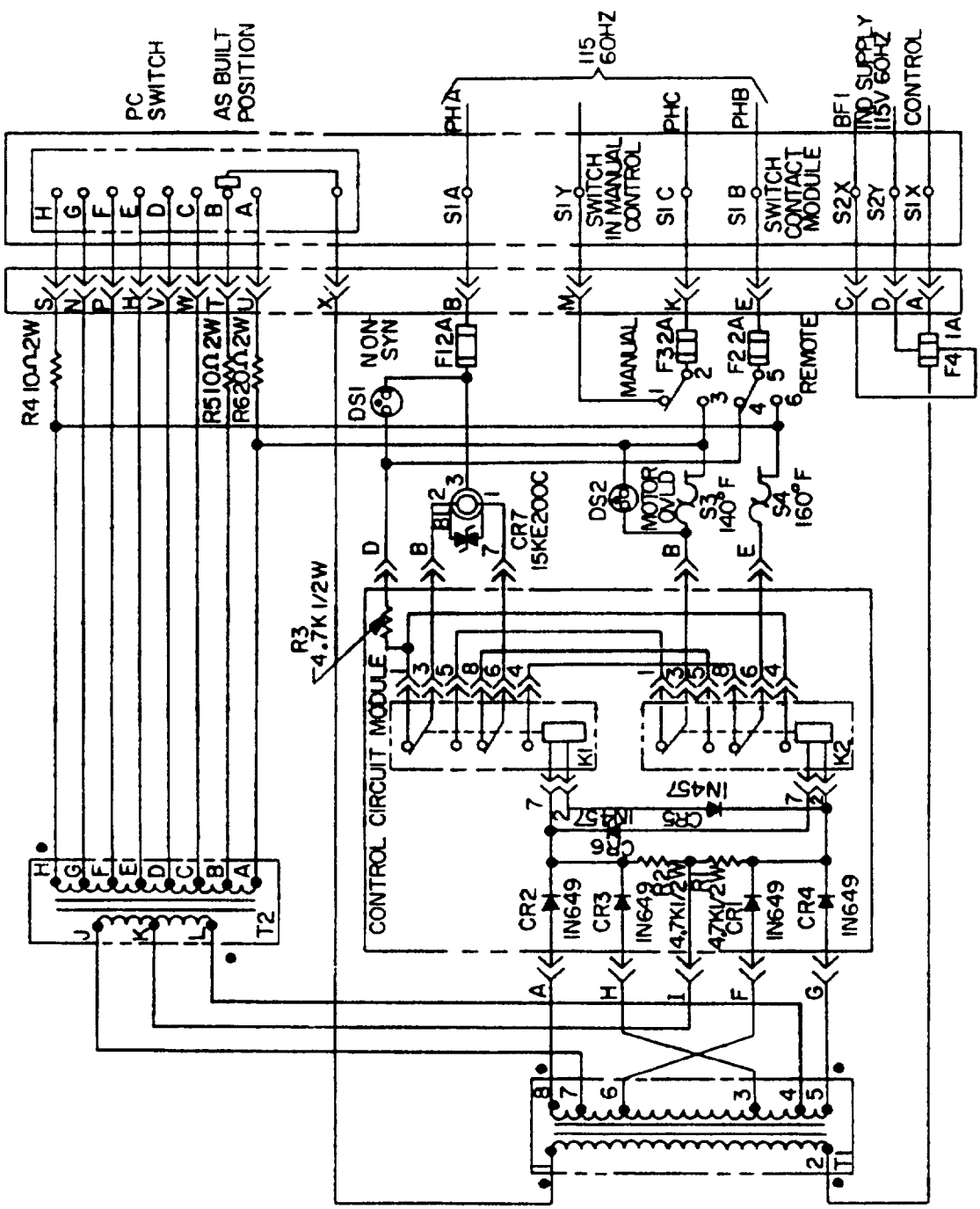


FIGURE 8. *Remote control circuit.*

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Repair parts: Repair parts shall be as specified in table VII.

TABLE VII. *Repair parts.*

Drawings 803-5727191(S3A4) piece No 815-5002992(S3A3) piece No		Description	Quantities required for quantity of switches installed					
			1–3	4–12	13–30	31–60	61–120	121 or more
Manual	Remote							
2	1	Enclosed switch, S3A4	1	1	1	1	2	2
(2992)	1	Enclosed switch, S3A3	1	1	1	1	2	2
66	66	Relay	2	2	2	2	2	4
29	48	Seal screw	8	8	8	8	8	16
93	26	Seal screw	14	14	14	14	14	28
36	45	Gasket	1	1	1	1	1	2
65	83	Gasket	2	2	2	2	2	4
158	158	Stop screw	1	1	1	2	2	3
32	32	Detent spring	1	1	1	2	3	6
	87	Toggle switch	1	1	1	2	3	4
	85	Indicator light	1	1	1	2	3	5
	86	Indicator light	1	1	1	2	3	5
		Lamp (M15098/11-001)	Note 2					
	10	Control circuit module	1	1	1	2	2	3
	93	Relay	1	2	3	Note 3		
	160	Washer, insul	2	2	2	4	6	8
	159	Spacer, insul	2	2	2	4	6	8
	81	Fuseholder	1	1	1	2	3	4
		Fuse carrier (1A0981)	1	1	2	3	4	5
	82	Fuse, 2.0 ampere	Note 4					
	83	Fuse, 1.0 ampere	Note 4					

NOTES
1. Refer to Drawing 803-4680149 for piece numbers
2. One hundred percent repair lamps shall be provided
3. Supply one for each 10 or fraction of 10 installed.
4. One hundred fifty percent repair fuses shall be supplied. Fuses shall be supplied in multiples of five, in standard commercial packages, with a minimum quantity of 15

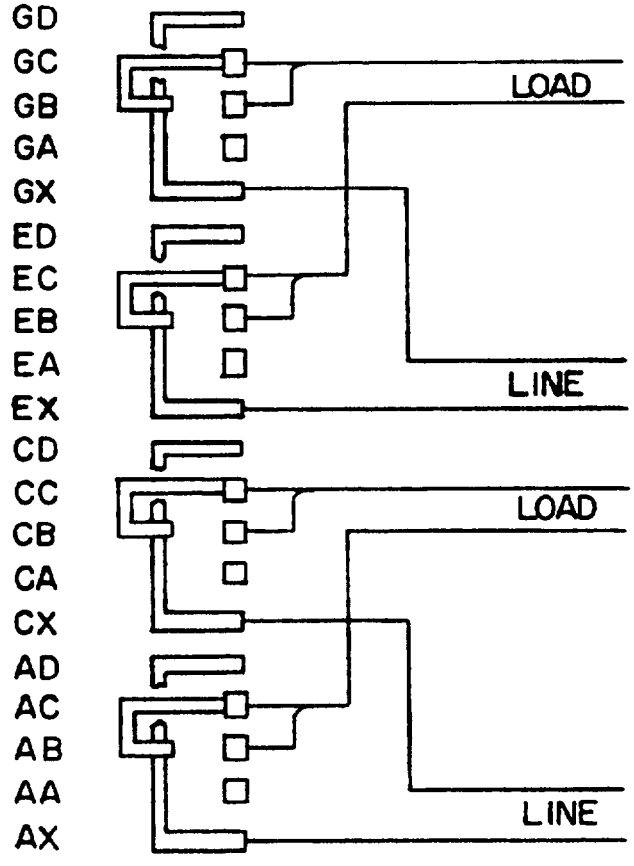
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Qualification samples: Two remote operated switch assemblies shall be submitted for qualification inspection.

NOTE: Manufacturers who have qualified to MIL-S-24187/4 are qualified by similarity.

Tests:

Loading of switches: Loading of switches for endurance and temperature rise tests of qualification and quality conformance, group C testing, shall be as shown on figure 9, and table VIII.



NOTE: Sections 1 and 2, row 1, shall be loaded at 2 amperes resistive at 15 volts dc. Sections 1 and 2, row 15, shall be loaded at 25 microamperes resistive at 1.5 volts.

FIGURE 9. *Test loading.*

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Temperature rise: Contacts to be monitored during temperature rise tests are specified in table VIII.

TABLE VIII. *Temperature measurements.*

Section	Row	Contacts
1	1	GX, GB, CX, CB
2	1	EX, EB

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5930-N681-09)

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